

WORK DELIVERABLES SUMMARY

Divyan Jain
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The following document details all software modules built and delivered across multiple phases of the project engagement. Each section outlines the specific features implemented and the technical work completed.

01. RTSP Streamer — Complete Rebuild & Feature Extension

Full rewrite of the EasyPlayer-based RTSP streaming app. All legacy dependencies were refactored or replaced. The rebuild introduced 4-screen simultaneous playback, hardware acceleration, and a drone telemetry OSD system powered by MAVLink/MAVSDK.

Category	Features Delivered
Primary Flight Display	• Dynamic Artificial Horizon with roll/pitch animation • Scrolling Speed Tape (left) and Altitude Tape (right) with center highlight • Vertical Speed Indicator (VSI) bar on altitude tape • Scrolling 360° Compass tape with N/E/S/W labels • Center crosshair with aircraft symbol
HUD Data Blocks	• Top-Left: Battery voltage, per-cell voltage, percentage (red below 20%) • Top-Right: ARM/DISARM state, GPS fix type, satellite count, flight timer • Bottom-Left: Live latitude & longitude • Bottom-Right: Throttle percentage, distance to home (Haversine formula)
Telemetry Backend	• MAVLink TCP listener for Wi-Fi/network drone connections • Auto-Home capture on ARM event • Auto flight timer (start on ARM, stop on DISARM) • Real-time distance-to-home calculation
Settings & OSD	• Individual toggles for Horizon, Tapes, Corner Info, Crosshair • Persistent settings via SharedPreferences • Instant-apply on returning to flight screen
Core App Features	• Live RTSP/HTTP stream playback via EasyPlayer engine • 4-screen simultaneous multi-feed grid layout • QR code scanner for stream URL input • Media manager: browse, view, and manage local recordings • Raw YUV frame export for computer vision use • In-app auto-updater with APK download and install prompt

Technical Overhaul	• AndroidX migration across all Java files and XML layouts • Removed conflicting gson-2.1.jar, resolved Gson version collision • AGP upgraded to 7.4.2 to fix D8 compiler NullPointerException • Java 8 desugaring enabled for MAVSDK v3.14.0 compatibility • minSdkVersion bumped to 24 (Android 7.0) per MAVSDK requirements
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02. HTTP Streamer — Built from Scratch

A brand-new Android streaming application built from the ground up using Google's Media3 ExoPlayer library. Supports HLS, DASH, and direct video files with a polished fullscreen UI.

Category	Features Delivered
Playback Engine	• Media3 ExoPlayer integration for HLS (.m3u8), DASH, and .mp4 • Error handling: invalid URL / lost connection recovery
UI & UX	• Immersive fullscreen (hidden status & navigation bars) • Landscape-locked orientation • Floating semi-transparent input overlay on video background • Auto-hiding menu and keyboard on playback start • Keep Screen On during active streams
Interactive Features	• Custom URL input field for any valid stream • Video quality selector — scans available resolutions, popup to force quality or Auto • Smart back navigation: stops video and returns to menu; second press exits

03. iNAV Configurator — Custom Android App

A purpose-built Android iNAV flight controller configurator with USB OTG support, implementing the MSP protocol from scratch for calibration and sensor monitoring.

Category	Features Delivered
Hardware & Connection	• USB OTG auto-detection via usb-serial-for-android driver • Connect/Disconnect toggle with color-coded status indicator (Red/Green)
MSP Protocol Engine	• Custom MSP packet encoder and decoder (Java ↔ byte array)

	• XOR checksum calculation and validation
Live Telemetry	• Background polling loop — UI refresh every 100ms • Real-time Accelerometer and Magnetometer X/Y/Z axis display • Stored pitch and roll trim offset display (MSP_ACC_TRIM)
Calibration Tools	• 6-point Accelerometer calibration (Flat, Upside Down, Left, Right, Nose Up, Nose Down) • Per-step visual feedback: yellow (sent) → green (confirmed by FC) • Compass calibration with 30-second guided countdown timer
System Controls	• Save & Reboot: writes calibration to EEPROM and restarts board • Factory Reset: zeroes acc trims, auto-saves and reboots

04. AndroidUSBCamera (AUSBC) — Modernization & Feature Addition

Modernized the AndroidUSBCamera open-source library and demo app with new interactive zoom features, build environment upgrades, and dependency migrations.

Category	Features Delivered
New Zoom Feature	• Pinch-to-Zoom: ScaleGestureDetector with linear 1.0x–5.0x range • Double-tap to instantly reset zoom to 1.0x • HUD zoom indicator overlay, auto-hides after 1.5 seconds • Matrix-based TextureView transformation for accurate magnification
Build Modernization	• compileSdk bumped to API 33, minSdk raised from 19 to 21 • Gradle wrapper updated to 8.5, AGP to 8.2.2, Kotlin plugin to 1.9.22 • Namespace declarations added to all sub-project Gradle files (AGP 8.x compliance) • BuildConfig generation compatibility configured in gradle.properties
Library Migrations	• Immersionbar: 3.0.0 → actively maintained 3.2.2 (geyifeng fork) • WebP Decoder: migrated to zjupure:webpdecoder 2.3.4.14.2 • Kotlin null-safety refactor across CameraActivity, CameraFragment, and widget views
Core Library Features	• Multi-road UVC camera support without system permissions • Multi-resolution preview (480p, 720p, 1080p+) • Image (.jpg) and video (.mp4) capture and recording • Raw H.264 / AAC / NV21 / RGBA stream access • OpenGL ES 2.0 live filter support • Audio capture from device mic or external camera mic (UAC)

*All modules listed above represent completed, functional deliverables.
Available for demo, code review, or APK testing upon request.*